Text-Augmented Semantic Feature Extraction and Difference Information Learning for Remote Sensing Image Change Captioning

Renlong Hang, *Member, IEEE*, Jinyu Luo, Hui Lin, and Qingshan Liu, *Senior Member, IEEE*

***Abstract*—Remote sensing image change captioning (RSICC) aims to generate sentence descriptions about land cover changes in bitemporal images. The effective acquisition of semantic-level change information is critical for this task. However, due to the effects of illumination interference, appearance similarities and scale differences between different objects, it is difficult to accurately extract change information from bitemporal images. In this article, we attempt to take advantage of the high-level semantic information inherent in text and propose a text-augmented semantic feature extraction and difference information learning model for RSICC. Specifically, we first predefine some text prompts for each remote sensing image and use the contrastive language-image pretraining (CLIP) model to select the most suitable text descriptions for them. Then, we adopt a refined segment anything model (SAM) to learn fine-grained visual features from each image, which is further enhanced via a designed selective text–image fusion (STIF) module. After that, to extract the semantic differences between bitemporal images, we propose a text-guided difference capture (TGDC) module capable of extracting multiscale difference information under the guidance of text differences between different-time images. Finally, a transformer-based caption generator is applied to generate sentence descriptions from the extracted difference information. In order to test the performance of our proposed model, we conduct comprehensive experiments on two widely used RSICC datasets, including LEVIR-CC and Dubai-CC. The experimental results show that our proposed model is able to outperform several state-of-the-art models, which validates the effectiveness of it. The codes of our proposed model will be released at https://github.com/Richardkimyo/TACC**

***Index Terms*—Change captioning, difference information learning, semantic feature extraction, text prompts.**

I. Introduction

THE swift progression of human civilization and the relentless evolution of the natural environment have precipitated unparalleled alterations to the Earth's surface. Identifying these transformations is of paramount importance for interdisciplinary research encompassing both the humanities and the natural sciences. Change detection is a widely used method for automatically detecting the changed information from bitemporal or multitemporal images captured over the same geographical area. It usually generates a binary map at the pixel level that describes whether each pixel changes or a semantic change map that highlights the types of change for each pixel [1]. Compared to change detection, change captioning is a more user-friendly method that uses sentence descriptions to interpret relationships and attributes of the changed area at the semantic level, thus attracting more and more attention in various fields, such as environmental assessment, urban planning, and disaster monitoring [2], [3].

In recent years, many works have been proposed for remote sensing image change captioning (RSICC), and they are generally composed of three stages, including visual feature extraction, difference information learning, and caption generation. For the extraction of visual features, convolutional neural networks and vision transformers [6] have become the main framework. For example, Cai et al. [7] proposed an interactive change-aware network, which uses ResNet [8] to extract visual features. Liu et al. [9] proposed a pure transformer model named a progressive scale-aware network, which uses a vision transformer to extract visual features. Following the feature extraction stage, difference information learning is often adopted to capture the changed information between bitemporal images. In [10], Chang et al. proposed an attentional encoder based on self-attention to accurately localize the variations between bitemporal images. In [4], Liu et al. decoupled the difference information learning task into two subtasks, which determine whether a change occurred and identify the specific changes that took place, respectively. Based on the changed information learned, the caption generation stage aims to transform it into sentence descriptions. To achieve this goal, Chouaf et al. [11] and Hoxha et al. [1] adopted recurrent neural networks (RNNs) as decoders to generate sentences sequentially. Recently, with the emergence of the transformer and its ability to process long-sequence sentences, the transformer decoder has begun to become the main choice for sentence generation [12], [13].

Most of the existing models can achieve satisfactory performance in simple scenes with clear semantics and limited environmental interference, but may encounter difficulties in complex scenes such as illumination interference, appearance similarities, and scale differences between different objects. Taking Fig. 1(a) as an example, strong illumination conditions obscure the object *house* in the blue box, making it less

Received 26 January 2025; revised 2 July 2025 and 16 July 2025; accepted 26 July 2025. Date of publication 4 August 2025; date of current version 15 August 2025. This work was supported in part by the Natural Science Foundation of China under Grant U21B2044 and Grant 62472230. *(Corresponding author: Qingshan Liu.)*

Renlong Hang and Hui Lin are with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: renlong_hang@163.com; huilin@nuist.edu.cn).

Jinyu Luo is with Tianchang Research Institute, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: jiny_luo@163.com).

Qingshan Liu is with the School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210023, China (e-mail: qsliu@njupt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3595422

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

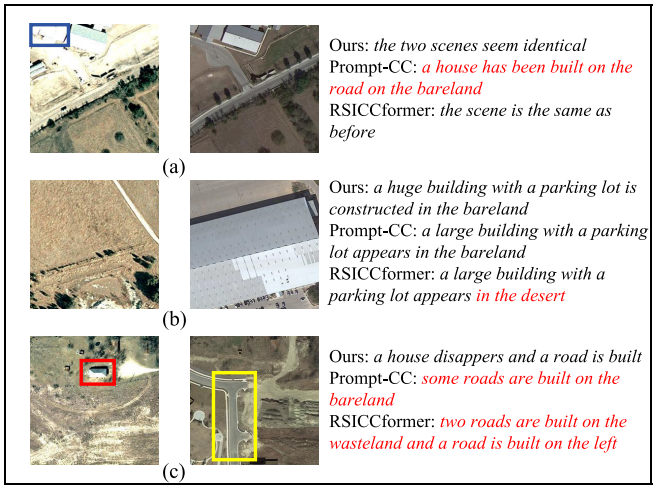


Fig. 1. Typical challenges in the task of RSICC. (a) Illumination interference. (b) Appearance similarities. (c) Scale difference. On the right side of each subfigure, the change captioning results acquired by Prompt-CC [4], RSICCformer [5] and our proposed model are provided. Red fonts indicate incorrect captions.

distinguishable from the surroundings. Therefore, the Prompt-CC model proposed in [4] mistakenly thinks that *a house has been built*. Another example is shown in Fig. 1(b) where *bare land* and *desert* have similar appearances. This similarity confuses the RSICCformer model proposed in [5], which misidentifies *bare land* as *desert*. In addition, another challenging case is the scale variations between different changed objects of interest. As demonstrated in Fig. 1(c), there exist two changed objects, including a small-scale object *house* in the red box and a large-scale object *road* in the yellow box. Both the Prompt-CC model and the RSICCformer model can capture the changed information about *road*, but overlook the *house*, which is not obvious as the *road*.

In order to address the aforementioned challenges, we attempt to make use of the high-level semantic information inherent in text and propose a text-augmented semantic feature extraction and difference information learning model for RSICC. Specifically, considering that there don't exist any text inputs for the task of RSICC, we first predefine some text prompts for each input image and use the contrastive language-image pretraining (CLIP) [14], [15] model to select the most suitable text descriptions for them. Then, we adopt FastSAM [16], a variant of the segment anything model (SAM) [17] with faster inference speed and low computational resource requirements to learn fine-grained visual features from each image. Such features are further enhanced by a selective text–image fusion (STIF) module, which is designed to explore the semantic information of object categories provided by text descriptions as clues to assist the model in extracting more comprehensive semantic features. After that, to learn the difference information between bitemporal images, we propose a text-guided difference capture (TGDC) module capable of extracting multiscale difference information under the guidance of text differences between different-temporal images. Finally, a transformer-based caption generator is used to generate sentences describing the content differences between bitemporal images based on the difference information extracted by the previous module. As shown on the right side of Fig. 1, our proposed model is capable of generating correct sentence descriptions in all three cases. The primary contributions of our proposed model can be summarized as follows.

1) Considering that the individual use of images is not sufficient enough to extract effective semantic features and difference information for RSICC in complex scenes, such as illumination interference and appearance similarities between different objects, we propose taking advantage of the rich semantic information provided by text. To achieve this goal, we predefine some text prompts and use CLIP to find the most suitable ones.
2) In order to make full use of the semantic information in text, we design an STIF module and a TGDC module during the feature extraction phase and the difference information learning phase, respectively. The STIF aims at using the global semantic feature extracted from text to enhance visual features. The purpose of the TGDC module is utilizing the text difference features to help learn multiscale difference information.
3) Comprehensive experiments are conducted on two widely used RSICC datasets, including LEVIR-CC [5] and Dubai-CC [1]. The quantitative and qualitative results demonstrate that our proposed model outperforms several state-of-the-art models.

The remainder of this article is organized as follows. In Section II, some related works about RSICC are introduced. In Section III, our proposed model is presented in detail. The experimental setup and results are discussed in Section IV. Finally, Section V provides a conclusion of this article.

II. Related Work

In this section, some previous research will be introduced, including remote sensing change detection and change captioning, both of which are related to our work.

A. Change Detection

Remote sensing change detection is a critical research area, focusing on identifying pixel-level differences across multitemporal remote sensing images. Existing methods can be broadly categorized into traditional approaches and deep learning techniques. Early traditional methods relied mainly on image processing algorithms, such as image differencing [18], image ratioing [19], and change vector analysis. Other widely used methods include principal component analysis and multivariate alteration detection. For example, Nielsen [20] improved the multivariate alteration detection method by implementing an iterative reweighting strategy that progressively emphasized observations with uncertain change statuses, thus enhancing its robustness. Classification-based methods have also been employed for change detection, which typically utilize classifiers to distinguish between changed and unchanged areas in a pixel-level manner. Popular classifiers include support vector machines, random forests, and Markov random fields. However, such methods are highly dependent on the quality of empirically designed features and have insufficient robustness under various situations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

Compared with traditional methods, deep neural networks have the ability to directly obtain high-level and robust features from raw data in an end-to-end manner. Therefore, many works have been devoted to improving the performance of change detection models using deep learning. However, in practical applications, since the resolution of historical data is usually low and there is a resolution difference between the currently acquired high-resolution images and historical images, this brings challenges to change detection. In order to solve the need for change detection between remote sensing images of different resolutions, Chen et al. [21] proposed a method called scale-invariant learning with implicit neural networks, which trains the model by synthesizing samples with random resolution differences, enabling it to perform change detection on bitemporal images with different resolution ratios. Many researchers also attempted to propose new fusion strategies to improve detection performance. Ye et al. [22] used the internal fusion property of 3-D convolution to simultaneously extract and fuse the feature information from bitemporal images. Then, they used the adjacent-level feature cross-fusion module they proposed to aggregate the complementary information between features of different convolutional layers to narrow the semantic gap between low-level features and high-level features. The outstanding performance of foundation models in downstream tasks has led some researchers to apply foundation models to change detection. Li et al. [23], proposed a general framework named the bitemporal adapter network, which aims to leverage the general knowledge accumulated in large-scale pretrained foundation models to reduce the dependence on task-specific labeled data and adapt it to change detection tasks. Zhang et al. [24] proposed SFCD-Net, a remote sensing change detection network that integrates SAM [17], a bitemporal feature interaction module, and a boundary loss function. In addition to research on model structure, some researchers have proposed new loss functions. Mall et al. [25] proposed a change-aware contrast loss function by combining short-term (seasonal) and long-term (structural) change modeling. This method dynamically detects long-term changes in images and adjusts the feature learning strategy only when significant changes are detected, thereby enhancing the sensitivity and robustness of feature representation. In summary, both change detection and change captioning aim to capture changes of interest within remote sensing images, but they differ fundamentally in how these changes are described. In this article, we focus on the task of change captioning.

B. Change Captioning

Change captioning aims to generate a natural language description of the changes observed between multitemporal images. The generated captions should be accurate, concise, and aligned with human perception. As a high-level semantic understanding task, change captioning has garnered increasing attention and witnessed promising advancements in recent years. Existing methods for change captioning typically consist of three key components: visual feature extraction, difference feature learning, and caption generation. As an initial step in processing images, visual feature extraction plays a crucial role in learning image content. Zhou et al. [26] proposed a single-stream extractor network based on ResNet [8] by contrastive pretraining on an additional remote sensing dataset, with the aim of improving the ability of the visual encoder to extract effective features from remote sensing images. In addition, the good generalization performance of foundation models in downstream tasks has led some researchers to use foundation models as visual feature extractors. For example, Zhu et al. [27] used SAM as a visual feature extractor and used LoRA [28] to fine-tune it for better adaptation to remote sensing images. Difference learning is a crucial step in change captioning methods as it directly impacts the performance of the difference description. Various strategies have been proposed to improve this step. For example, Li et al. [29] proposed the symmetric difference transformer, which improves the ability to represent change by aligning the difference features obtained in different ways. Tu et al. [30] proposed a cross-semantic relation measurement block. This approach draws inspiration from the concept of long short-term memory networks, using it to measure the correlation between the difference features and the bitemporal images, thereby enhancing the model's ability to identify subtle differences. Liu et al. [31] introduced the state space model (SSM) [32] and Mamba [33] into RSICC and proposed a spatial difference-aware SSM to improve the spatial change perception capability. Using the temporal scanning feature of Mamba, they proposed a temporal traversing SSM to scan the bitemporal features in a time-crossing manner to enhance the model's temporal understanding and information interaction. In the caption generation stage, the methods used in most works are relatively fixed, basically using RNN and Transformer decoder as caption generators. However, there are exceptions. Liu et al. [4] employed GPT-2 [34] as a caption generator through the prompt learning method.

Although some progress has been made in RSICC, the performance of existing methods is often degraded when complex scenes are encountered, such as illumination interference and high appearance similarities between different objects, because these factors make it difficult for models to accurately learn and distinguish between objects in ambiguous semantics. In addition, the diversity in target object scales further complicates change recognition, as small target changes are frequently overlooked, leading to incomplete or inaccurate descriptions of scene changes. Addressing these issues requires more robust semantic extraction and multiscale feature learning techniques. To address the shortcomings of previous methods, we propose generating a text prompt for each remote sensing image. This text prompt serves as an auxiliary clue to enhance the model's semantic understanding of the image and difference information learning, thereby improving the accuracy of the generated captions.

III. Methodology

A. Overall Architecture

Fig. 2 shows the flowchart of our proposed model. Assume $\{\mathbf{I}^1, \mathbf{I}^2\} \in \mathbb{R}^{3\times H\times W}$ represents an input pair of bitemporal images, where H and W represent the height and width of each image, respectively. We first define some text prompts for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

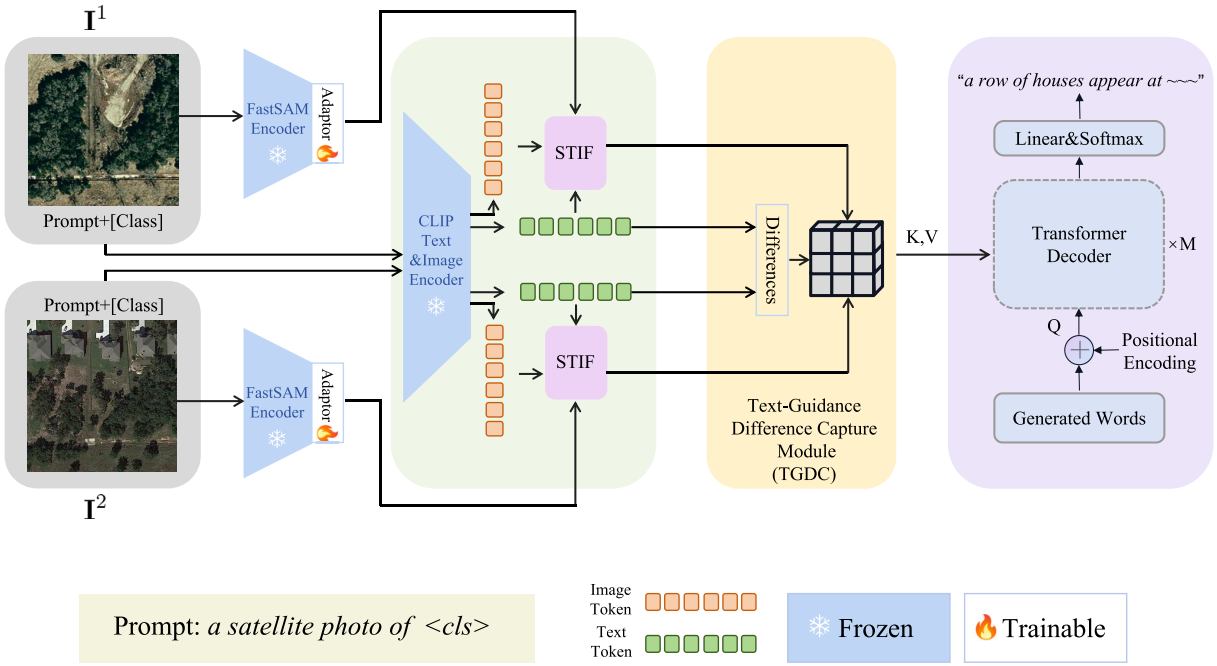


Fig. 2. Overall framework of our proposed model. For a pair of input images, a predefined text prompt is first provided. Then, FastSAM is used to extract multiscale features from them, which are enhanced by STIF. Such enhanced features are fed into TGDC to learn multiscale difference information. Finally, a transformer decoder is applied to generate a sentence description about the difference.

TABLE I
CATEGORY NAMES OF SOME COMMONLY ENCOUNTERED OBJECTS IN THE FIELD OF REMOTE SENSING

bare land	paved road	building	aircraft
lake	river	sea	parking area
forest	residential area	bridge	vegetation
harbor	island	sports complex	industrial area
tree	desert	vehicle	airport
park	field	square	commercial area

$\mathbf{I}^j$, $j \in \{1, 2\}$ and use CLIP [14] to encode its global image and text representations. Then, inspired by [35], an adaptor refined FastSAM is applied to extract fine-grained visual features with four different scales from $\mathbf{I}^j$, which can be denoted as $\mathbf{F}_i^j$ where $i \in \{1, 2, 3, 4\}$. The spatial size of $\mathbf{F}_i^j$ is $H/2^{(i+1)} \times W/2^{(i+1)}$. After that, we design an STIF module to enhance $\mathbf{F}_i^j$ by selectively combining the visual and text features, and acquire an enhanced feature $\mathbf{M}_i^j$. Next, we propose a TGDC module to learn multiscale difference information from each pair of $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$ with the guidance of text differences. Finally, the difference information is fed into a transformer decoder to generate change captions.

B. Text Prompt

The core idea of our proposed model is taking advantage of the high-level semantic information from text as a guide to extract semantic features and difference information. However, there are no text descriptions about input images for the RSICC task. Therefore, it is necessary to first define some text prompts for each input image. To this end, we adopt a hand-crafted prompt "*a satellite photograph of* ⟨**cls**⟩" where ⟨**cls**⟩ represents a predefined category name. Taking inspiration from the work in [36], we also define a total of 24 object categories commonly encountered in the field of remote sensing image processing. The detailed 24-category names are shown in Table I. For the lth category, we can generate a corresponding text prompt $\mathbf{T}_l, l \in \{1, 2, \ldots, 24\}$. Taking the first category *bare land* as an example, its corresponding prompt $\mathbf{T}_1$ is "*a satellite photograph of bare land*." It serves as an additional semantic cue, telling the model *what* the content of the image is.

C. Selective Text And Image Fusion (STIF)

Considering that there often exist different categories of objects in $\mathbf{I}^j$, the predefined text prompt with only one category name in Section III-B is not sufficient enough to represent the semantic information of it. An intuitive method to address this issue is to combine multiple suitable categories. As shown in Fig. 3, for each image $\mathbf{I}^j$ and each text prompt $\mathbf{T}_l$, we apply CLIP to extract image tokens and text tokens, and calculate their similarity. After that, the text prompts with top-k highest similarities are selected. The categories contained within these selected prompts are then combined together to generate a new text prompt $\mathbf{T}_{\text{new}}^j$, which can be described as "*a satellite photograph of* ⟨$\mathbf{cls}_1$⟩ *and* ⟨$\mathbf{cls}_2$⟩ *and* $\cdots$ *and* ⟨$\mathbf{cls}_k$⟩," where $\{\mathbf{cls}_1, \ldots, \mathbf{cls}_k\}$ are the corresponding k selected category names. Such a regenerated text prompt provides a more comprehensive representation of $\mathbf{I}^j$.

After obtaining $\mathbf{T}_{\text{new}}^j$, we use the CLIP text encoder to extract a text feature from it, which is then employed to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

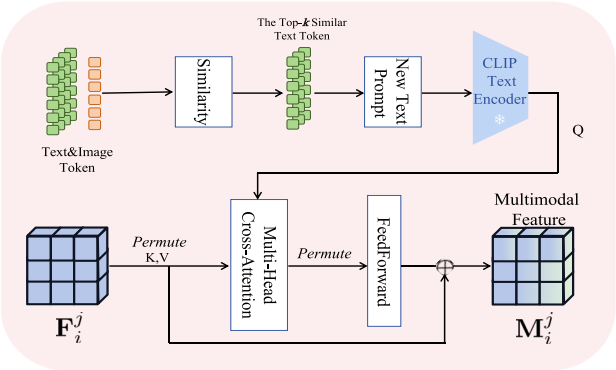


Fig. 3. Detailed structure of the STIF module. The top-k text prompts are first selected and the category names contained in them are combined together to regenerate a text prompt. Such prompt is fed into CLIP to extract a text feature which is used to enhance the visual feature via a multihead cross-attention (MHCA) operator.

enhance $\mathbf{F}_i^j$ via an MHCA operator. The detailed process can be formulated as

$$\mathbf{E}_{\text{new}}^j = \text{CLIP}_{\text{text}}\left(\mathbf{T}_{\text{new}}^j\right) \tag{1}$$

$$\mathbf{M}_i^j = \text{MHCA}\left(\mathbf{E}_{\text{new}}^j, \mathbf{F}_i^j, \mathbf{F}_i^j\right) \tag{2}$$

$$\mathbf{M}_i^j = \text{FeedForward}\left(\mathbf{M}_i^j\right) + \mathbf{F}_i^j \tag{3}$$

where $\text{CLIP}_{\text{text}}$ is the text encoder of CLIP, the FeedForward represents an operator which consists of two linear layers with an RELU activation function, $\mathbf{E}_{\text{new}}^j$ represents the extracted text feature by CLIP, and $\mathbf{M}_i^j$ is the enhanced feature of $\mathbf{F}_i^j$. MHCA contains multiple parallel single-head cross-attention and can be formulated as

$$\text{MHCA} = \text{Concat}(\text{head}_1, \ldots, \text{head}_{\text{h}})\text{W}^O \tag{4}$$

$$\text{head}_{\text{c}} = \text{CrossAtt}\left(\text{W}_{\text{c}}^Q\mathbf{E}_{\text{new}}^j, \text{W}_{\text{c}}^K\mathbf{F}_i^j, \text{W}_{\text{c}}^V\mathbf{F}_i^j\right) \tag{5}$$

$$\text{CrossAtt} = \text{Softmax}\left(\frac{\text{W}_{\text{c}}^Q\mathbf{E}_{\text{new}}^j(\text{W}_{\text{c}}^K\mathbf{F}_i^j)^{\text{T}}}{\sqrt{d}}\right)\text{W}_{\text{c}}^V\mathbf{F}_i^j \tag{6}$$

where W_{c}^Q, W_{c}^K, W_{c}^V, and W^O are trainable weight matrices, d is the output dimension, and Softmax is the activation function.

D. Text-Guided Difference Capture

Difference information learning is critical for RSICC. Most of current models only focus on exploring visual features to achieve this goal, which may easily contain irrelevant change information. Here, we propose a text-guided difference information learning module named TGDC to alleviate this issue. As shown in Fig. 4, TGDC captures the differences from both text and visual perspectives. For a pair of text prompts $\mathbf{T}_{\text{new}}^1$ and $\mathbf{T}_{\text{new}}^2$, we first derive a difference prompt from them by combining the different category names. For example, if $\mathbf{T}_{\text{new}}^1$ is "*a satellite photograph of lake and parking area and tree*" and $\mathbf{T}_{\text{new}}^2$ is "*a satellite photograph of residential area and parking area and tree*," the difference prompt would be "*a satellite photograph of lake and residential area.*" Such prompt is then fed into the CLIP text encoder to extract a difference text feature $\mathbf{T}^{\text{dif}}$.

In parallel to the extraction of difference text feature, the visual features $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$ extracted in Section III-C are

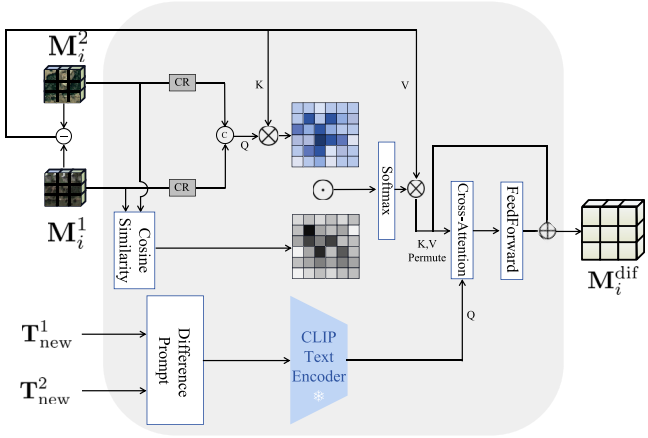


Fig. 4. Detailed structure of the TGDC module. CR represents channel reduction. The difference of text prompts between bitemporal images is fed into the CLIP text encoder to learn difference information, which can be used to improve the learning of difference visual features.

used to acquire a difference visual feature. Specifically, $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$ are subtracted to obtain a preliminary difference feature $\mathbf{M}_i^{\text{dif}}$. $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$ are compressed to half of the original channel numbers, and concatenated along the channel dimension to obtain $\mathbf{M}_i^{\text{cat}}$. Then, $\mathbf{M}_i^{\text{dif}}$ as a query and $\mathbf{M}_i^{\text{cat}}$ as a key are multiplied to obtain $\mathbf{Sim}_i$. By measuring the correlation between the difference features and the bitemporal features, it decouples the common/difference features from the image pair. The above process can be formulated as

$$\mathbf{M}_i^{\text{dif}} = \mathbf{M}_i^2 - \mathbf{M}_i^1 \tag{7}$$

$$\mathbf{M}_i^2 = \text{ChannelReduction}\left(\mathbf{M}_i^2\right) \tag{8}$$

$$\mathbf{M}_i^1 = \text{ChannelReduction}\left(\mathbf{M}_i^1\right) \tag{9}$$

$$\text{Q}_i^{\text{dif}} = \text{W}_{\text{dif}}^Q\mathbf{M}_i^{\text{dif}} \tag{10}$$

$$\text{K}_i^{\text{cat}} = \text{W}_{\text{cat}}^K\text{Concat}\left(\mathbf{M}_i^1, \mathbf{M}_i^2\right) \tag{11}$$

$$\text{V}_i^{\text{cat}} = \text{W}_{\text{cat}}^V\text{Concat}\left(\mathbf{M}_i^1, \mathbf{M}_i^2\right) \tag{12}$$

$$\mathbf{Sim}_i = \frac{\text{Q}_i^{\text{dif}}\left(\text{K}_i^{\text{cat}}\right)^{\text{T}}}{\sqrt{d}} \tag{13}$$

where ChannelReduction is a 1×1 convolution operator, Concat is a concatenation operator, W_{dif}^Q, W_{cat}^K, and W_{cat}^V are trainable weight matrices, and d is the dimension of Q_i^{dif} and K_i^{cat}. Furthermore, we also calculate a similarity matrix $\mathbf{Cos_i}$ by computing a cosine similarity between $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$. In $\mathbf{Cos_i}$, its values would be very small when the corresponding regions change significantly and vice versa. Based on $\mathbf{Sim}_i$ and $\mathbf{Cos_i}$, the difference visual feature $\mathbf{M}_i^{\text{dif}}$ can be derived as

$$\mathbf{Sim}_i = \text{Softmax}(\mathbf{Sim}_i \odot (\mathbf{1} - \mathbf{Cos}_i)) \tag{14}$$

$$\mathbf{M}_i^{\text{dif}} = \text{V}_i^{\text{cat}}\mathbf{Sim}_i \tag{15}$$

where $\odot$ represents the Hadamard product.

After obtaining $\mathbf{M}_i^{\text{dif}}$,$\mathbf{T}^{\text{dif}}$ serves as a guide to constrain it learn change information of interest in a targeted manner, which can be denoted as

$$\mathbf{M}_i^{\text{dif}} = \text{FeedForward}(\text{MHCA}(\mathbf{T}^{\text{dif}}, \mathbf{M}_i^{\text{dif}}, \mathbf{M}_i^{\text{dif}})) + \mathbf{M}_i^{\text{dif}}. \tag{16}$$

Since the changed objects are with different scales, we input $\mathbf{M}_1^{\text{dif}}$, $\mathbf{M}_2^{\text{dif}}$, $\mathbf{M}_3^{\text{dif}}$, and $\mathbf{M}_4^{\text{dif}}$ into a feature pyramid network (FPN) [37] to learn multiscale difference information $\mathbf{M}_{\text{fus}}^{\text{dif}}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

E. Caption Decoder

To generate an accurate sentence description about the change information, we use a standard Transformer decoder as a caption decoder. The decoder consists of multiple transformer decoder layers, each of which comprises multihead masked self-attention, encoder–decoder attention, and the FeedForward network. The encoder–decoder attention is essentially MHCA, which is used to complete the interaction between the encoder and decoder. In our model, we first fuse the four scale features in $\mathbf{M}_i^1$ and $\mathbf{M}_i^2$, respectively, through FPN, and concatenate these two fused features with $\mathbf{M}_{\text{fus}}^{\text{dif}}$ together along the channel dimension to obtain the final visual feature $\mathbf{X}_{\text{fus}}$. Then, we use encoder–decoder attention to achieve visual and text relationship modeling, where Q from the previous masked self-attention sublayer, and K,V are from the transformation of $\mathbf{X}_{\text{fus}}$. Mathematically, the output of the lth cross-attention sublayer can be expressed as

$$\mathbf{S}_l^{\text{text-img}} = \text{MHCA}(\mathbf{S}_l, \mathbf{X}_{\text{fus}}, \mathbf{X}_{\text{fus}}) \tag{17}$$

where $\mathbf{S}_l$ is the sentence embedding from the masked self-attention sublayer. The feature $\mathbf{S}_l^{\text{text-img}}$ is subsequently processed by a feedforward network. After N layers of transformer decoder, the retrieved word embedding is fed to a linear layer followed by a softmax activation function to obtain the word probabilities, denoted by $\mathbf{P} = \left[\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \ldots, \hat{\mathbf{p}}_n\right] \in \mathbb{R}^{n \times m}$, where n is the length of the generated caption and m is the size of the vocabulary. The model is trained by minimizing the cross-entropy loss between the predicted word probabilities and the reference captions. The loss function is defined as

$$\mathbf{L} = -\sum_{s}^{n}\sum_{l}^{m} \mathbf{p}_{s,l} \log \hat{\mathbf{p}}_{s,l} \tag{18}$$

where $\mathbf{p}_{s,l}$ is the one-hot encoding and $\hat{\mathbf{p}}_{s,l}$ is the predicted probability of the sth word in the reference caption at step l.

IV. Experiments

A. Data Description

1) LEVIR-CC: The first RSICC dataset we conduct experiments on is LEVIR-CC [5]. It comprises 10077 pairs of high-resolution bitemporal images, each with a spatial size of 256 × 256 pixels and a resolution of 0.5 m/pixel. These images, derived from the LEVIR-CD [39] dataset, capture aerial views from 20 different regions in Texas, USA, with a time span of 5 to 14 years between each pair. This dataset is annotated with 50 385 descriptive captions, with each image pair associated with five sentences that describe either the presence or absence of changes. Among the image pairs, 5038 pairs depict significant changes, such as the construction of new buildings or alterations in land use, while the remaining 5039 pairs show no notable changes. The vocabulary used in the dataset is drawn from annotated sentences, filtered to include only words that appear at least five times, resulting in a lexicon of 497 unique words. In order to train and evaluate our proposed model, we follow the standard data partitioning scheme and divide the whole dataset into training, validation, and test sets, consisting of 6815, 1333, and 1929 image pairs, respectively.

2) DUBAI-CC: The second RSICC dataset is Dubai-CC [1], which is used to support the analysis and description of urbanization changes in the Dubai region. The bitemporal images in Dubai-CC are captured by the Enhanced Thematic Mapper Plus (ETM+) sensor aboard Landsat 7 on May 19, 2000 and June 16, 2010. It is cropped into tiles with 50 × 50 pixels, resulting in 500 bitemporal tiles. To capture semantic-level changes, each tile is annotated with five natural language descriptions, referencing Google Maps and publicly available sources. In total, the dataset provides 2500 captions, with a maximum length of 23 words and an average of 7.35 words per caption. The vocabulary spans 266 unique words. Changes are categorized into six main types, including roads, houses, buildings, green areas, lakes, and islands. For model development and evaluation, the dataset is divided into training, validation, and testing subsets containing 300, 50, and 150 tiles, respectively.

B. Experimental Setup

1) Evaluation Metrics: To quantitatively evaluate the performance of different change captioning models, we choose four principal metrics, including BLEU-N [40] (where N ranges from 1 to 4), METEOR [41], ROUGE-L [42], and CIDEr-D [43]. BLEU-N assesses the precision of n-grams in the generated sentences compared to ground-truth sentences, emphasizing the fidelity of the generated text to the reference sentences at various levels of granularity. ROUGE-L focuses on the recall rate, measuring the longest common subsequences between the generated and reference sentences, thus prioritizing the coverage of content from the ground truths. METEOR goes a step further by accounting for semantic accuracy and fluency, introducing a penalty for less fluent sentences, thereby ensuring not only the presence of correct words but also their appropriate usage in context. CIDEr-D, tailored specifically for image captioning tasks, leverages term frequency-inverse document frequency (TF-IDF) scores to provide differentiated weights to n-grams, focusing on the uniqueness and relevance of the descriptive terms utilized. They are chosen for their proven correlation with human judgment, ensuring that higher scores on these metrics reflect a closer alignment with human perceptions of quality and accuracy in describing the changes observed between bitemporal or similar images. Besides, we follow the previous work [4] to adopt an overall metric S_m^* to integrate the above metrics. It is defined as

$$S_m^* = \frac{1}{4} * (\text{BLEU-4} + \text{METEOR} + \text{ROUGE}_L + \text{CIDEr-}D). \tag{19}$$

2) Implementation Details: We implement all models based on the PyTorch framework and use the ViT-B/32 of CLIP to obtain text and image tokens. All models are trained and evaluated on the NVIDIA RTX 4090 GPUs. In the training stage, we use the Adam optimizer [44]. We set the maximum epoch to 50. The initial learning rate is 0.0001. We evaluate the model on the validation set after each training epoch. The learning rate will decay by a weight decay of 0.7 after 7 epochs. The model achieving the highest BLEU-4 score on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PERFORMANCE OF DIFFERENT MODELS ON THE LEVIR-CC DATASET

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
DUDA	81.44	72.22	64.24	57.79	37.15	71.04	124.32	72.58
MCCFormers-S	79.90	70.26	62.68	56.68	36.17	69.46	120.39	70.68
MCCFormers-D	80.42	70.87	62.86	56.38	37.29	70.32	124.44	72.11
Diffusion-RSCC	-	-	-	60.90	37.80	71.50	125.60	73.95
PSNet	83.86	75.13	67.89	62.11	38.80	73.60	132.62	75.46
RSICCformer	84.72	76.27	68.87	62.77	39.61	74.12	134.12	77.66
Prompt-CC	83.66	75.73	69.10	63.54	38.82	73.72	136.44	78.13
Ours	**85.49**	**77.41**	**70.52**	**64.62**	**40.07**	**74.96**	**137.17**	**79.21**
▲	+0.77	+1.14	+1.42	+1.08	+0.46	+0.84	+0.73	+1.08

TABLE III
PERFORMANCE OF DIFFERENT MODELS ON THE DUBAI-CC DATASET

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
DUDA	58.85	43.59	33.63	25.39	22.05	48.34	62.78	39.64
MCCFormers-S	52.97	37.02	27.62	22.57	18.64	43.29	53.81	34.58
MCCFormers-D	64.65	50.45	39.36	29.48	25.09	51.27	66.51	43.09
PSNet	59.64	47.95	37.89	29.93	25.11	52.57	67.97	43.90
RSICCformer	63.95	50.38	39.30	30.66	25.37	51.36	68.28	43.91
Diffusion-RSCC	-	-	-	33.30	27.40	56.50	88.70	51.48
Ours	**72.17**	**58.65**	**48.24**	**38.59**	**28.17**	**58.93**	**95.45**	**55.28**
▲	+7.52	+8.20	+8.88	+5.29	+0.77	+2.43	+6.75	+3.80

the validation data over the 50 epochs is selected as optimal to be used for evaluation on the test set.

C. Comparisons With State-of-the-Art Models

To verify the performance of our proposed model, we compare it with several state-of-the-art models, including Capt-Rep-Dif [45], Capt-Att [45], Capt-Dual-Att [45], DUDA [45], MCCFormer-S [46], MCCFormer-D [46], Diffusion-RSCC [47], RSICCformer [5], Prompt-CC [4], and PSNet [9]. As shown in Table II, compared to the second-best model Prompt-CC, our proposed model improves BLEU-4, METEOR, ROUGE-L, and CIDEr-D by 1.08%, 1.25%, 1.24%, and 0.73%, respectively, with the overall metric S_m^* increasing by 1.08%. In Table III, compared to the second-best model Diffusion-RSCC, our proposed model improves BLEU-4, METEOR, ROUGE-L, and CIDEr-D by 5.29%, 0.77%, 2.43%, and 6.75%, respectively, with the overall metric S_m^* increasing by 3.80%. These results indicate that the captions generated by our proposed model are more similar to the reference sentences, and the similarities and differences between the images are well captured. The comparison results show that our proposed model exhibits excellent performance in the RSICC task, which is due to our designed text-augmented method for RSICC. With the support of text description, the model is able to extract the semantics of changing scenes at a deeper level, thereby correctly modeling visual information and captioning.

Fig. 5 compares the change captioning results achieved by our proposed model and three best comparison models, including Prompt-CC [4], RSICCformer [5] and Diffusion-RSCC [38]. In this figure, five ground-truth sentences are presented for each instance, and red fonts are used to indicate incorrect captions. As shown in Fig. 5(a), the newly added objects include a building and a parking lot, which are difficult to distinguish based on their appearance, making Prompt-CC and RSICCformer recognize only one of them. In Fig. 5(b), the newly added buildings are too similar to the ground in color and fuzzy in texture, so it is difficult to judge whether these buildings exist. RSICCformer is confused by its high appearance similarity and does not recognize the real changes. Compared with these two models, although the appearance similarities between the changing object and the surroundings are very high, our proposed model still accurately identifies them. Fig. 5(c) and (d) highlights that the illumination interference often causes pseudo-changes. RSICCformer and Diffusion-RSCC mistakenly interpret the scene as changing ones. Compared with them, our proposed model demonstrates

TABLE IV
COMPUTATIONAL OVERHEAD AND PERFORMANCE OF DIFFERENT MODELS

Method	**Params (M)**	**Inference (S)**	S_m^*
RSICCformer	56.20	0.048	77.66
Prompt-CC	193.36	0.146	78.13
PSNet	93.11	0.054	75.46
MCCFormer-D	135	0.050	72.11
Ours	94.14	1.06	79.21

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

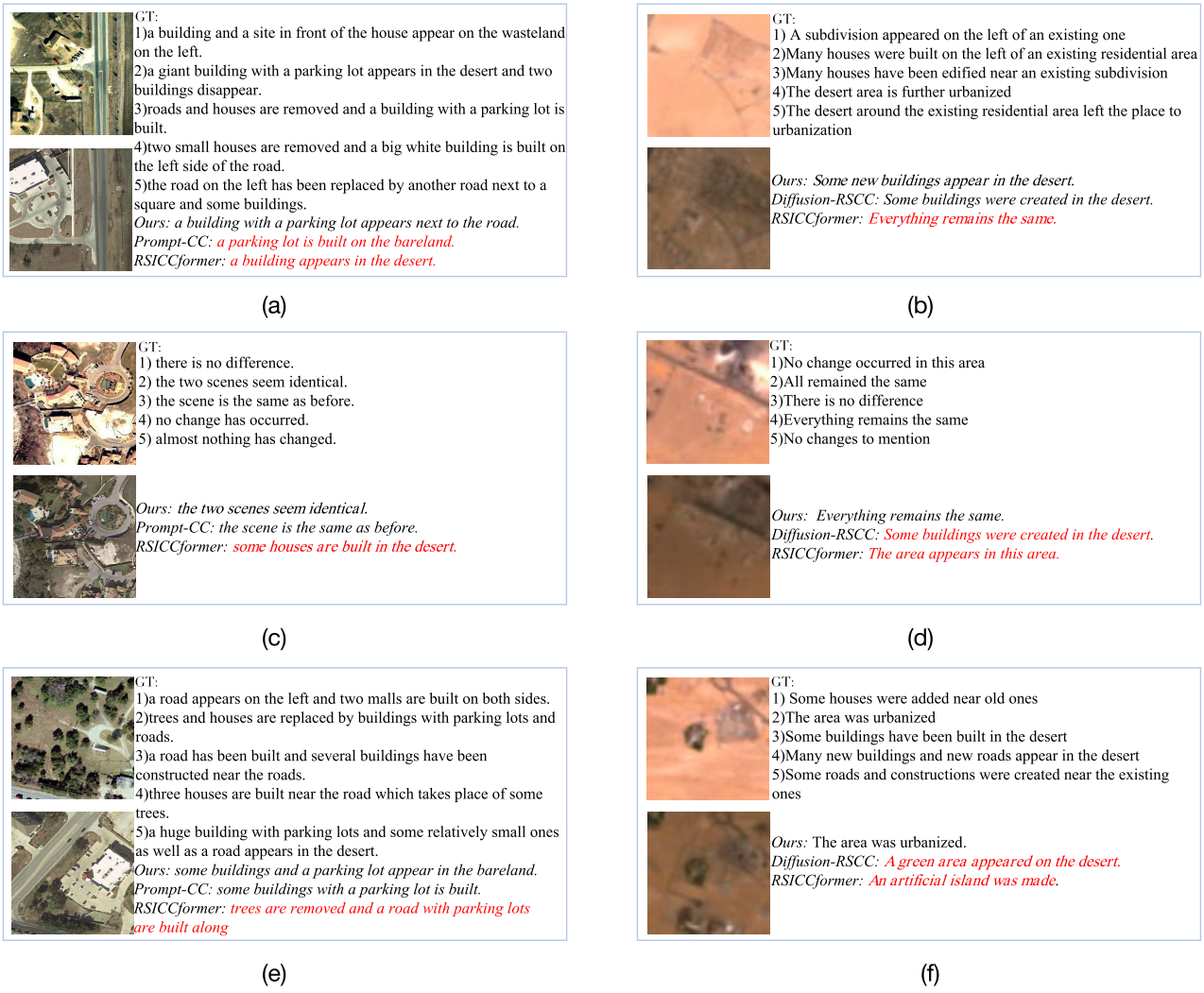


Fig. 5. Change captioning results achieved by our proposed model and three best comparison models (i.e., Prompt-CC [4], RSICCformer [5], and Diffusion-RSCC [38]) under different conditions. (a) and (b) Under high appearance similarities. (c) and (d) Under illumination interference. (e) and (f) Under multi-scale changes. In these figures, GT represents five ground-truth captions, and red fonts indicate incorrect captions. The left three examples are from the LEVIR-CC dataset, and the right ones are from the Dubai-CC dataset.

TABLE V
ABLATION STUDY OF STIF AND TGDC MODULES

Dataset	STIF	TGDC	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
LEVIR-CC			83.40	74.81	67.29	61.24	38.11	73.76	132.39	76.38
	✓		83.87	75.52	68.67	62.91	38.23	72.57	134.21	76.98
		✓	84.67	76.14	69.23	63.96	39.01	73.56	134.36	77.72
	✓	✓	**85.49**	**77.41**	**70.52**	**64.62**	**40.07**	**74.96**	**137.17**	**79.21**
Dubai-CC			69.41	55.28	44.74	36.37	26.87	54.40	91.55	52.29
	✓		71.45	57.30	46.54	37.31	26.51	54.65	88.90	51.84
		✓	71.86	57.75	46.70	37.64	27.01	57.08	92.54	53.56
	✓	✓	**72.17**	**58.65**	**48.24**	**38.59**	**28.17**	**58.93**	**95.45**	**55.28**

strong robustness in handling such scenarios. Fig. 5(e) and (f) shows the results of different models in the case of scale diversity of changing objects. Our proposed model can still recognize them accurately.

In addition, we compare the computational overhead of our proposed model with four typical models which have open-source codes and achieve better performance than the other comparison models. Table IV shows detailed results of different models in terms of training parameters, inference time, and S_m^*. It can be observed that our proposed model has fewer training parameters than the second best model Prompt-CC in terms of S_m^*, because the parameters in CLIP is frozen. However, during the inference phase, CLIP takes much time to compute, making the inference time of our proposed model longer than the other models. Nevertheless, our model can achieve the best BLEU-4 value than the other models.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF COSINE SIMILARITY AND DIFFERENCE PROMPT IN TGDC

Dataset	Cosine Similarity	Difference Prompt	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
LEVIR-CC			83.31	75.26	67.97	63.04	38.26	73.11	133.41	76.96
	✓		84.31	76.04	69.37	63.92	38.99	73.62	134.17	77.68
		✓	84.26	75.79	68.97	63.77	38.67	73.08	133.89	77.35
	✓	✓	**85.49**	**77.41**	**70.52**	**64.62**	**40.07**	**74.96**	**137.17**	**79.21**
Dubai-CC			71.30	57.78	46.71	36.55	26.49	55.67	82.79	50.37
	✓		70.80	57.31	46.87	37.23	27.03	58.41	92.78	53.86
		✓	71.64	57.77	47.43	37.92	27.86	57.21	91.80	53.69
	✓	✓	**72.17**	**58.65**	**48.24**	**38.59**	**28.17**	**58.93**	**95.45**	**55.28**

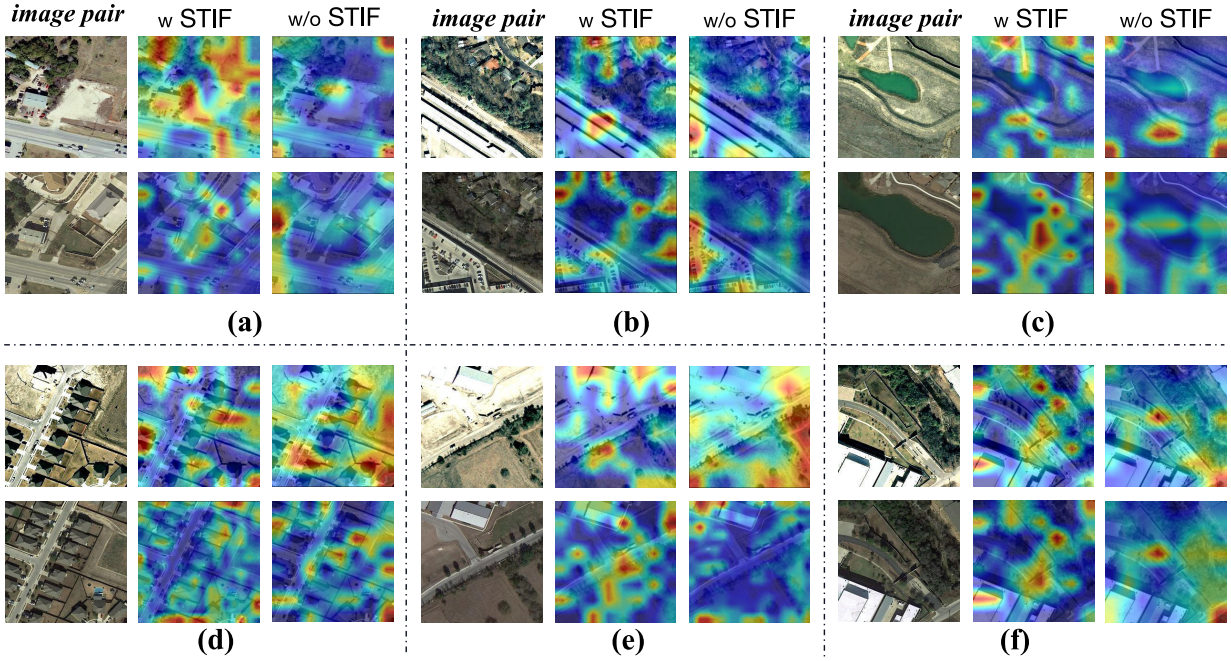


Fig. 6. Visualization results of our proposed model with (w) and without (w/o) using STIF under different conditions. (a)–(c) Under high appearance similarities. (d)–(f) Under illumination interference.

D. Ablation Study

1) Effects of STIF and TGDC Modules: STIF and TGDC serve as the core components of our proposed model, playing an important role in the process of semantic feature extraction and difference information learning. To demonstrate their effectiveness, we conduct a series of ablation experiments. The results are presented in Table V, where "✓" indicates that the corresponding module is used. At first, we construct a *baseline* model by removing both modules and leaving FastSAM for visual feature extraction and FPN for multiscale feature fusion. Then, we add STIF and TGDC sequentially to the *baseline* model to evaluate their contributions. Several conclusions can be drawn. First, when STIF is added to the *baseline* model, the overall performance is improved from 76.38 to 76.98 on LEVIR-CC, because it can leverage the direct interaction between images and text, guiding the model to perform a more accurate extraction of semantic features. Second, when TGDC is added to the *baseline* model, the overall performance can also be increased on both datasets. More importantly, when both modules are used, our proposed model achieves the best performance, with an overall metric improving from 76.38 to 79.21 on the LEVIR-CC dataset, and from 52.29 to 55.28 on the Dubai-CC dataset.

Fig. 6 shows the attention maps with or without STIF in some instances, where "w" and "w/o" indicate whether STIF is incorporated into our proposed model. The top three figures show the results of STIF under a condition of high appearance similarities, and the bottom three figures show the performance under a condition of illumination interference. For example, in Fig. 6(a), a building with a color almost identical to the ground surface appears in the image above. With STIF, the model successfully detects it, while without STIF, the model ignores its existence and mistakenly mixes the building with the ground. In Fig. 6(d), illumination interference makes it difficult to determine whether the building in the upper left corner is an existing building or a newly built one. With the integration of the STIF module, the model is successfully aware that the building already exists, and the correct extraction of semantic features provides support for the subsequent learning of difference information. These examples demonstrate the effectiveness of STIF in enhancing the model's ability to extract semantic features under complex visual conditions.

Fig. 7 visualizes the attention maps with or without TGDC, where "w" and "w/o" represent whether TGDC is adopted in our proposed model. We can clearly observe that when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
COMPARISON BETWEEN FASTSAM AND RESNET-34

Dataset	Extractor	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
LEVIR-CC	ResNet-34	83.55	75.59	69.07	63.41	38.62	73.11	136.06	77.80
	FastSAM	**85.49**	**77.41**	**70.52**	**64.62**	**40.07**	**74.96**	**137.17**	**79.21**
Dubai-CC	ResNet-34	69.59	57.43	43.21	35.23	25.27	54.31	89.73	51.13
	FastSAM	**72.17**	**58.65**	**48.24**	**38.59**	**28.17**	**58.93**	**95.45**	**55.28**

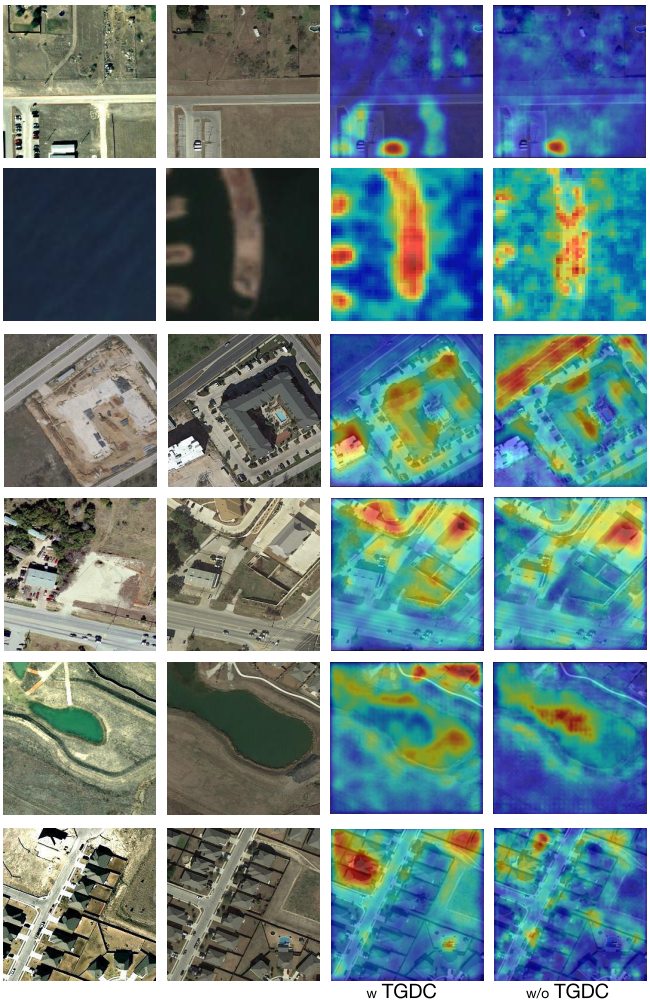


Fig. 7. Visualization results of our proposed model with (w) and without (w/o) using TGDC.

TGDC is used, the model can better focus on the various changes that we are interested in. Taking the first row of Fig. 7 as an example, it highlights the model's ability to handle multiscale changes, including small-scale vehicle changes and large-scale road changes. When the TGDC module is used, the model demonstrates greater sensitivity to changes on different scales, effectively addressing the challenge posed by scale diversity and extreme scale variations. In contrast, without the TGDC module, the model struggles to detect such changes comprehensively. The fourth and fifth rows of Fig. 7 reflect that under the condition of high appearance similarities, our method still has excellent difference recognition ability. Following that, the sixth row of Fig. 7 further illustrates the model's robustness in scenarios with illumination interference. Although the building in the upper left corner is difficult to observe, our model still accurately recognizes that the house exists from the beginning and is not a new building. The sixth row of Fig. 7 also indicates that although STIF may pay attention to some objects with irrelevant changes shown in Fig. 6(d), the significant difference learning ability of TGDC will filter out these irrelevant information. These examples highlight the model's capacity to recognize various changes and maintain high sensitivity across diverse scales and complex semantic environments.

2) Effects of Cosine Similarity and Difference Prompt: The cosine similarity and the difference prompt are components within the TGDC, both designed to enhance the model's ability to identify changes. To verify their effectiveness, we perform additional ablation experiments. As shown in Table VI, the performance of our proposed model is higher when both the cosine similarity and the difference prompt are used in TGDC compared to cases when only one of them is used. This improvement occurs because cosine similarity and difference prompt have distinct yet complementary roles. The cosine similarity provides a reference for visual differences for TGDC. The difference prompt identifies semantic differences in the text prompts of the image pair, helping TGDC better detect the changes of interest because it provides textual cues describing the differences in categories. In summary, these two components contribute significantly to improving the ability to learn information about differences in different aspects.

3) Comparison between FastSAM and ResNet: Compared to the traditional approach of employing two Siamese convolutional networks for temporal feature extraction, we take inspiration from previous work [35] and adopt FastSAM as our feature extractor. To evaluate its effectiveness, we compare FastSAM with ResNet-34, which has a similar parameter size. ResNet-34 requires 81 MB, while FastSAM requires 68 MB. As presented in Table VII, FastSAM shows significant performance improvements over ResNet-34 despite having fewer parameters. This improvement can be attributed to FastSAM's superior fine-grained feature extraction capabilities. Furthermore, to address the inductive biases introduced by training models on natural images, we use the adaption method described in [35]. This approach ensures that the model maintains strong generalization ability on RSICC datasets.

E. Quantitative Analysis on k Value

In the STIF module, $\boldsymbol{k}$ is a key hyperparameter that determines the number of categories in the text prompt. It is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PERFORMANCE OF DIFFERENT k VALUES ON LEVIR-CC AND DUBAI-CC DATASETS

	LEVIR-CC									Dubai-CC							
k	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*	k	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr-D	S_m^*
1	84.66	76.83	69.74	63.77	38.96	73.78	136.42	78.23	1	69.42	56.69	46.90	**38.73**	27.57	57.78	92.75	54.21
2	84.20	75.29	68.25	62.97	38.67	73.57	134.07	77.32	2	**72.86**	57.40	46.04	36.31	25.91	55.52	82.32	50.02
3	85.35	77.00	69.96	64.34	39.12	74.57	136.86	78.72	3	72.17	58.65	**48.24**	38.59	**28.17**	**58.93**	**95.45**	**55.28**
4	**85.57**	**77.53**	70.29	64.58	39.96	74.61	**137.24**	79.10	4	71.93	**58.93**	46.24	37.48	27.96	56.79	85.13	51.84
5	85.49	77.41	**70.52**	**64.62**	**40.07**	**74.96**	137.17	**79.21**	5	69.68	56.89	47.12	36.29	26.17	55.50	79.74	49.43
6	83.37	74.95	67.46	62.07	38.59	73.24	136.01	77.48	6	70.17	55.77	45.47	36.70	27.36	54.60	79.55	49.55
7	85.33	77.08	70.11	64.32	39.62	74.61	136.82	78.84	7	70.15	56.62	46.04	36.34	25.53	55.27	87.94	51.27
8	85.57	77.09	70.21	64.51	39.89	74.63	137.13	79.04	8	72.63	57.22	45.74	35.83	25.36	54.80	81.87	49.47
9	84.51	76.88	69.82	63.91	39.07	73.63	136.53	78.29	9	66.56	52.51	42.52	34.71	22.37	51.21	79.20	46.87

important for our proposed model to learn sufficient but not redundant semantic information. So, we conduct experiments to test its effects on the final performance. As shown in Table VIII, we change $\boldsymbol{k}$ from 1 to 9 with a step of 1 on LEVIR-CC, and when $\boldsymbol{k}$ equals 5, our proposed model achieves the best overall performance. When it is greater than 5, our proposed model shows a trend of first decreasing, then increasing, and decreasing again. Similar phenomenons can be observed on Dubai-CC and the best performance is achieved when $\boldsymbol{k}$ equals to 3. These results reflect that too many or too few category texts are not conducive to the model. When the image is larger and contains more types of objects, the value of $\boldsymbol{k}$ will increase accordingly, and vice versa. The value of $\boldsymbol{k}$ is also related to the recognition of differences. The difference prompt we set up in TGDC is designed to help the model capture interesting changes through the differences between text prompts.

V. CONCLUSION

In this article, we proposed a text-augmented semantic feature extraction and difference information learning model for RSICC. It first constructed text prompts for input images. Such prompts were used to enhance the visual features extracted by a variant SAM model (i.e., FastSAM) via our designed STIF module. Then, the prompts' difference between a pair of bitemporal images was employed to guide the learning of difference information, making our proposed TGDC module capture semantic-level differences from both text and visual perspectives. Finally, the difference information was converted into sentence descriptions with a transformer decoder. In order to test the performance of our proposed model, we conducted experiments on two widely used RSICC datasets, including LEVIR-CC and Dubai-CC. In addition, we implemented many ablation experiments to evaluate the effectiveness of each module. Although our proposed model achieved better performance than the comparison models, it still has some limitations. First, TGDC currently lacks the ability to judge whether a text prompt is wrong. This is extremely important to ensure the robustness of the module. Second, our method assumes the availability of class names or prompts, which may not generalize to more complex or unstructured environments. In the future work, we plan to develop reliability-aware mechanisms to filter or correct inaccurate textual guidance, and extend our framework to handle open-vocabulary issues.

REFERENCES

[1] G. Hoxha, S. Chouaf, F. Melgani, and Y. Smara, "Change captioning: A new paradigm for multitemporal remote sensing image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627414.

[2] S. H. Khan, X. He, F. Porikli, and M. Bennamoun, "Forest change detection in incomplete satellite images with deep neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5407–5423, Sep. 2017.

[3] J. Gao, L. Zhao, and X. Li, "NWPU-MOC: A benchmark for fine-grained multicategory object counting in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5606614.

[4] C. Liu et al., "A decoupling paradigm with prompt learning for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5622018.

[5] C. Liu, R. Zhao, H. Chen, Z. Zou, and Z. Shi, "Remote sensing image change captioning with dual-branch transformers: A new method and a large scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633502.

[6] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[7] C. Cai, Y. Wang, and K.-H. Yap, "Interactive change-aware transformer network for remote sensing image change captioning," *Remote Sens.*, vol. 15, no. 23, p. 5611, Dec. 2023.

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2016, pp. 770–778.

[9] C. Liu, J. Yang, Z. Qi, Z. Zou, and Z. Shi, "Progressive scale-aware network for remote sensing image change captioning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jun. 2023, pp. 6668–6671.

[10] S. Chang and P. Ghamisi, "Changes to captions: An attentive network for remote sensing change captioning," *IEEE Trans. Image Process.*, vol. 32, pp. 6047–6060, 2023.

[11] S. Chouaf, G. Hoxha, Y. Smara, and F. Melgani, "Captioning changes in bi-temporal remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 2891–2894.

[12] D. Sun, Y. Bao, J. Liu, and X. Cao, "A lightweight sparse focus transformer for remote sensing image change captioning," 2024, *arXiv:2405.06598*.

[13] X. Li, B. Sun, Z. Wu, S. Li, and H. Guo, "CD4C: Change detection for remote sensing image change captioning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 9181–9194, 2025.

[14] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[15] L. Fan, D. Krishnan, P. Isola, D. Katabi, and Y. Tian, "Improving CLIP training with language rewrites," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 35544–35575.

[16] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[17] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[18] J. Gao et al., "Infrared image change detection of substation equipment in power system using Markov random field," in *Proc. Int. Conf. Comput. Intell. Inf. Syst. (CIIS)*, 2017, pp. 332–337.

[19] M. N. Sumaiya and R. Shantha Selva Kumari, "Logarithmic mean-based thresholding for SAR image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1726–1728, Nov. 2016.

[20] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi{-} and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.

[21] H. Chen et al., "Continuous cross-resolution remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623320.
[22] Y. Ye et al., "Adjacent-level feature cross-fusion with 3-D CNN for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5618214.
[23] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.
[24] D. Zhang et al., "Integrating SAM with feature interaction for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4513011.
[25] U. Mall, B. Hariharan, and K. Bala, "Change-aware sampling and contrastive learning for satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5261–5270.
[26] Q. Zhou, J. Gao, Y. Yuan, and Q. Wang, "Single-stream extractor network with contrastive pre-training for remote-sensing change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624514.
[27] Y. Zhu et al., "Semantic-CC: Boosting remote sensing image change captioning via foundational knowledge and semantic guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5648916.
[28] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.
[29] Y. Li, X. Zhang, X. Cheng, P. Chen, and L. Jiao, "Intertemporal interaction and symmetric difference learning for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5641413.
[30] Y. Tu et al., "Semantic relation-aware difference representation learning for change captioning," in *Proc. Findings Assoc. Comput. Linguistics, ACL-IJCNLP*, 2021, pp. 63–73.
[31] C. Liu, K. Chen, B. Chen, H. Zhang, Z. Zou, and Z. Shi, "RSCaMa: Remote sensing image change captioning with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[32] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.
[33] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[34] A. Radford et al., "Improving language understanding by generative pre-training," OpenAI, San Francisco, CA, USA, Tech. Rep., 2018.
[35] L. Ding et al., "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.
[36] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.
[37] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.
[38] X. Yu, Y. Li, and J. Ma, "Diffusion-RSCC: Diffusion probabilistic model for change captioning in remote sensing images," 2024, *arXiv:2405.12875*.
[39] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[40] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "Bleu: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. meeting Assoc. for Comput. Linguistics*, 2002, pp. 311–318.
[41] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. ACL Workshop Intrinsic Extrinsic Eval. Measures Mach. Transl. Summarization*, 2005, pp. 65–72.
[42] C.-Y. Lin, "Rouge: A package for automatic evaluation of summaries," in *Proc. Workshop Text Summarization ACL*, Barcelona, Spain, 2004, pp. 74–81.
[43] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4566–4575.
[44] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[45] D. H. Park, T. Darrell, and A. Rohrbach, "Robust change captioning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4624–4633.
[46] Y. Qiu et al., "Describing and localizing multiple changes with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1971–1980.
[47] X. Yu, Y. Li, J. Ma, C. Li, and H. Wu, "Diffusion-RSCC: Diffusion probabilistic model for change captioning in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5617013.



Renlong Hang (Member, IEEE) received the Ph.D. degree in meteorological information technology from Nanjing University of Information Science and Technology, Nanjing, China, in 2017.

From 2018 to 2019, he was a Post-Doctoral Researcher with the Department of Computer Science and Electrical Engineering, University of Missouri–Kansas City, Kansas City, MO, USA. He is currently a Professor with the School of Computer and Software, Nanjing University of Information Science and Technology. He has authored or co-authored more than 50 peer-reviewed articles in international journals, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. His research interests include machine learning, pattern recognition, and their applications to remote sensing image processing.



Jinyu Luo received the B.E. degree in computer science and technology from the School of Science and Information, Qingdao Agricultural University, Qingdao, China, in 2023. He is currently pursuing the M.S. degree in computer technology with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing, China.

His research interests include deep learning and remote sensing image processing.



Hui Lin received the M.S. degree in computer science and technology from Nanjing University of Information Science and Technology, Nanjing, China, in 2024, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision, deep learning, and their applications to remote sensing and meteorology.



Qingshan Liu (Senior Member, IEEE) received the Ph.D. degree in pattern recognition and intelligence systems from Chinese Academy of Sciences, Beijing, China, in 2003.

After the doctoral studies, he joined the National Laboratory of Pattern Recognition, Chinese Academy of Sciences. From 2004 to 2005, he was an Associate Researcher with the Multimedia Laboratory, The Chinese University of Hong Kong, Hong Kong. From 2006 to 2011, he worked with the Department of Computer Science, Computational Biomedicine Imaging and Modeling Center, Rutgers University, New Brunswick, NJ, USA, and The State University of New Jersey, Piscataway, NJ, USA. From 2011 to 2023, he was a Professor with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing. He is currently a Professor with the School of Computer Science, Nanjing University of Posts and Telecommunications. His research interests include pattern recognition and image and video understanding.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:34:50 UTC from IEEE Xplore. Restrictions apply.